AutoShotV2: Calibrated Classification and Post-Processing for Short-Video Shot Boundary Detection

Senh Manh Tang, Tien Manh Do, Thai Son Tran, and Hoai Viet Vo

Abstract—**Shot boundary detection (SBD) is a fundamental preprocessing step for video indexing, retrieval, summarization, and automated editing. Modern short-video platforms introduce a challenging operating regime characterized by brief video durations, dense edits, visually diverse transitions, and sparse boundary frames. This paper presents AutoShotV2, a lightweight SBD refinement framework. Instead of repeating computationally expensive neural architecture search or updating the full backbone, AutoShotV2 freezes the searched feature extractor and trains a compact classification head on cached features. The resulting logits are calibrated via temperature scaling, smoothed with a one-dimensional Gaussian filter, and thresholded using a fixed operating point selected before test evaluation. In our primary deployment configuration, AutoShotV2 achieves an F1 score of 0.8545 on the 200-video SHOT test set, outperforming the reproduced AutoShot baseline by 1.40 percentage points. The deployed model also reaches high performance on BBC documentary footage (0.9656), though performance challenges persist on open-world ClipShots (0.7529) due to domain shift. To isolate individual component contributions, we conduct a controlled ablation study under a frozen validation-only protocol. The analysis indicates that most of the performance gain is attributable to post-processing, while alternative training losses do not add consistent gains. Uncertainty quantification via paired bootstrap intervals supports positive deltas on SHOT and ClipShots, and a training-seed study observes F1 standard deviation below 0.5 percentage points on every dataset. Finally, we provide validation manifests, per-video statistics, and checksummed artifacts and scripts.**

Index Terms—**Shot boundary detection, short video, AutoShot, frozen features, focal loss, probability calibration, temperature scaling, Gaussian smoothing, video understanding.**

I. INTRODUCTION

Effectively parsing video content is a fundamental requirement for many downstream applications such as video indexing, content retrieval, summarization, and automated editing. Within this pipeline, shot boundary detection (SBD) serves as a critical preliminary step: it partitions a video into temporally coherent shots by identifying frames where camera views, scene contents, or editing states transition [1], [2]. Classical methods based on hand-crafted cues such as color histograms and motion differences are efficient but fragile under fast camera motion, flashes, and modern editing styles [3].

Deep architectures such as TransNet and TransNetV2 [5], [6] learn spatiotemporal representations directly from frame sequences and have advanced modern SBD systems. Concurrently, the rise of short-video platforms (e.g., TikTok, YouTube Shorts) has introduced a new, challenging operating regime: brief videos, dense editing, and heavy special effects. To address it, AutoShot [4] leverages neural architecture search (NAS) to select a 3-D convolutional backbone, demonstrating notable performance on the newly introduced SHOT dataset. Further improving these models, however, encounters a practical bottleneck: the NAS pipeline requires substantial computation for supernet training and architecture evaluation. Moreover, short-video SBD suffers from extreme class imbalance — boundary frames are exceptionally rare — and complex visual effects that act as hard negatives.

We study an alternative refinement path that separates the frozen representation from the decision stage. Instead of re-running the NAS process or updating the searched AutoShot backbone, AutoShotV2 trains a compact classification head on cached 4864-dimensional features. This frozen-feature setting isolates a decision-stage design space: binary cross-entropy, Focal Loss [9], and boundary-aware many-hot supervision can be compared without changing the representation. Because SBD decisions are made by thresholding scores, the decision stage also includes probability calibration and temporal smoothing. Temperature scaling [10] rescales logits without changing their ranking, and a one-dimensional Gaussian filter suppresses isolated peaks caused by flashes, camera motion, or short-lived visual effects.

With a fixed deployment configuration, AutoShotV2 achieves an F1 score of 0.8545 on the SHOT test set, corresponding to an improvement of 1.40 percentage points over the reproduced AutoShot baseline. For comparison, a supplemental checkpoint evaluated with oracle per-dataset thresholds reaches an F1 score of 0.8607 on SHOT (see Section IV-D). A controlled replication under a frozen, validation-only five-fold protocol separates training choices from post-processing choices: the training-loss alternatives change F1 only marginally, whereas calibrated post-processing adds 1.62 percentage points on SHOT and 4.57 percentage points on ClipShots in F1 over the BCE control. Paired bootstrap intervals support the positive deltas on SHOT and ClipShots, but not on BBC.

In summary, our main contributions are:

- We propose AutoShotV2, an efficient SBD pipeline that improves short-video boundary detection without the computational burden of updating the NAS-searched

S. M. Tang and T. M. Do are with the Faculty of Information Technology, University of Science, Vietnam National University Ho Chi Minh City, Vietnam. T. S. Tran supervised this research. H. V. Vo is with the Faculty of Information Technology, University of Science, Vietnam National University Ho Chi Minh City, Vietnam.

backbone, reaching an F1 score of 0.8545 on SHOT with a fixed deployment configuration.

- We evaluate compact head variants and a loss-design space, including Focal Loss and boundary-aware many-hot supervision, and show that most of the performance gain is attributable to calibrated post-processing, while alternative training losses do not add consistent gains.
- We define a validation-only, source-stratified selection protocol that fixes temperature, smoothing, and threshold before test evaluation, and we quantify uncertainty with paired video bootstrap intervals and calibration diagnostics including adaptive and class-balanced ECE.
- We report experiments on SHOT, BBC, and ClipShots with cache identity checks, split/run manifests, a three-seed robustness study, and checksum-based artifact packaging.

This paper employs two complementary evaluation tracks. The deployed checkpoint supplies the headline numbers and is preserved through tracked, checksummed result files. A controlled replication with full logit caches anchors the logit-level confidence intervals, calibration diagnostics, ablations, and seed study (Section IV-B).

II. Related Work

Shot boundary detection. Shot boundary detection has long been a central task in video processing. Early systems detect frame-to-frame discontinuities with hand-crafted cues such as color histograms, edge-change ratios, motion compensation, and thresholding heuristics. These methods are computationally simple, but they often confuse lighting changes, camera motion, or object movement with actual boundaries [3]. Survey studies group the main challenges around abrupt cuts, gradual transitions, illumination changes, object and camera motion, and dataset-dependent evaluation protocols [1], [2]. These failure modes are amplified in short videos, where edits are dense and effects can be visually similar to true transitions.

Deep SBD models. Deep SBD models learn temporal patterns that are difficult to encode manually [13], [14], ranging from hierarchical clustering for broadcast reuse [20] to large-scale spatiotemporal CNNs [21]. TransNet casts SBD as dense frame-level prediction over low-resolution frame sequences [6]. TransNetV2 extends this design with dilated 3-D convolutions, frame-similarity features, color histograms, and dual prediction heads for single-frame and all-transition labels [5]. These models provide fast and accurate baselines, but their decision quality still depends on thresholded frame-level scores, making calibration and temporal stability important for deployment.

AutoShot and short-video SBD. Most established SBD datasets, such as BBC, RAI, and ClipShots, emphasize documentaries, broadcast footage, or longer user-generated videos. AutoShot addresses the short-video setting by introducing SHOT, a dataset with brief clips, dense editing, vertical layouts, complex gradual transitions, and heavy visual effects [4]. AutoShot uses NAS [8], [11], [12] to search over approximately 83.9 million candidate architectures composed of 3-D separable convolution blocks and temporal Transformer blocks [22]. The search trains a single-path one-shot supernet [23] and explores architectures with Bayesian optimization rather than differentiable relaxation [24]. Although the searched backbone improves SHOT performance, repeating the search and retraining pipeline is computationally costly. AutoShotV2 therefore asks how much can be gained by keeping this representation frozen and refining only the decision stage.

Imbalance, calibration, and temporal smoothing. Frame-level SBD is severely imbalanced because boundary frames are rare. Focal Loss reduces the contribution of easy examples and is a natural alternative to BCE [9], while soft and neighborhood-expanded targets such as label smoothing [25] motivate the boundary-aware many-hot supervision explored in this paper. SBD is also threshold-sensitive: temperature scaling rescales logits without changing their ranking [10], while richer alternatives such as Platt scaling [15] and isotonic regression [16] require more calibration data. Measuring calibration is itself nontrivial under sparse positives, because equal-width ECE can be dominated by the majority class; this motivates adaptive and class-conditional estimators [17]. AutoShotV2 evaluates these mechanisms separately so that improvements from training losses and post-processing are not conflated.

Efficient adaptation of frozen representations. Reusing a pretrained backbone without updating it is a common strategy, from linear probing of frozen features to parameter-efficient modules such as adapters [18] and low-rank updates [19]. Our setting is deliberately conservative: the searched AutoShot representation is fully frozen, a small classification head is trained on cached features, and the final adaptation happens through validation-selected calibration and smoothing. This keeps adaptation inexpensive and auditable, while also exposing the limits of a frozen representation under the ClipShots domain shift.

III. Methodology

This section formulates the shot boundary detection problem, summarizes the TransNetV2 and AutoShot components needed to define the frozen backbone, and presents the AutoShotV2 decision-stage pipeline built on top of it. Note that the convolutional blocks and search spaces are inherited from prior work; AutoShotV2 modifies only the decision stage described from Section III-F onward.

A. Problem Formulation

Given a video represented as an ordered sequence of T frames $\{x_t\}_{t=1}^{T}$, SBD predicts a binary sequence $\{b_t\}_{t=1}^{T}$, where $b_t{=}1$ denotes a shot boundary (an abrupt cut or a gradual transition) at frame t. A model outputs a boundary probability sequence $\{p_t\}_{t=1}^{T}$ with $p_t \in [0,1]$, and the final boundary set is obtained by thresholding:

$$\hat{b}_t = \begin{cases} 1, & p_t > \theta, \\ 0, & p_t \leq \theta, \end{cases} \tag{1}$$

where θ is a fixed operating point selected before test evaluation. Following standard practice, a predicted boundary counts as correct when it falls within 2 frames of a ground-truth boundary, and consecutive positive predictions are grouped into one transition.

B. Spatial-Temporal Factorized Convolution

Full 3-D convolutional kernels of size $k \times k \times k$ are expensive on frame sequences. The TransNetV2 family therefore factorizes each 3-D convolution into a 2-D spatial convolution ($1{\times}k{\times}k$, learning edges, textures, and color regions per frame) followed by a 1-D temporal convolution ($k \times 1 \times 1$, capturing motion patterns along the temporal axis). With C input/output channels and k=3, the parameter count drops from $27C^2$ to $9C^2 + 3C^2 = 12C^2$, a reduction of roughly 56% that also imposes a useful inductive bias separating spatial content from temporal motion.

C. Dilated DDCNN Blocks

The backbone stacks Dilated Convolutional Neural Network (DDCNN) blocks. Each block splits its input features into four parallel temporal-convolution branches with exponentially increasing dilation rates $d \in \{1, 2, 4, 8\}$, concatenates the branch outputs along the channel dimension, and applies batch normalization with ReLU activation. For temporal kernel size K=3, the cumulative receptive field after block k grows as

$$R_k = R_{k-1} + 2\,(K-1) \sum_{d\in\{1,2,4,8\}} d \;=\; R_{k-1} + 60, \tag{2}$$

with R_0=1. Six stacked blocks yield a theoretical receptive field of up to 361 frames; since inference uses sliding windows of 100 frames, the effective context is capped by the window but remains wide enough for long dissolves and fades.

D. Similarity Branch

A parallel similarity branch complements the convolutional features with two forms of visual consistency: (i) a 512-bin RGB histogram per frame, whose cosine distances between consecutive frames yield a 101-dimensional color-shift descriptor, and (ii) learned similarity, the cosine similarity between normalized deep features from intermediate convolutional layers. Both are fused through a fully connected layer with 128 units and ReLU activation, and the result is concatenated with the convolutional features before classification.

E. AutoShot Backbone Search

While TransNetV2 fixes a configuration of identical DDCNN blocks, AutoShot searches the backbone configuration for short-form video with neural architecture search [4], [8], [11], [12]. The space covers four DDCNN block variants (differing in channel configuration and spatial feature sharing) across six layers plus a choice of zero to four Transformer layers [22] at the seventh, about $83.9{\times}10^6$ candidate architectures in total. The search trains a single-path one-shot supernet [23] explored with Bayesian optimization (differentiable alternatives such as DARTS [24] were considered by the original authors), and the selected model is refined with knowledge distillation and weight grafting. The architecture found under the F1 criterion uses DDCNNV2 blocks of 4–12 filters with shared spatial convolutions in layers 2–4 and no Transformer layer. Its penultimate stage exposes a 4864-dimensional feature vector, which we freeze and cache:

$$h_t \in \mathbb{R}^{4864}. \tag{3}$$

All AutoShotV2 components operate on h_t without updating the backbone.

AutoShotV2 improves the decision stage of SBD: it trains a new classification head on the frozen features of Eq. (3) and adds probability calibration and temporal smoothing before thresholding. Fig. 1 shows the deployed inference pipeline.

F. Classification Head Architecture

The head maps each cached feature vector h_t to a shared hidden state

$$u_t = \mathrm{Dropout}\big(\mathrm{ReLU}(W_f h_t + c_f)\big), \tag{4}$$

with $W_f \in \mathbb{R}^{1024\times4864}$ and dropout probability 0.5. Two parallel linear branches then produce logits

$$z_t^{(1)} = W_1 u_t + c_1, \qquad z_t^{(2)} = W_2 u_t + c_2. \tag{5}$$

The primary branch $z_t^{(1)}$ is trained against one-hot boundary labels and is the only branch used at inference. The auxiliary branch $z_t^{(2)}$ can be supervised with boundary-aware many-hot targets to supply denser gradients around each transition. Fig. 2 illustrates the head.

G. Training Objectives as a Design Space

Boundary frames are exceptionally rare ($< 1\%$ of frames), so the training loss is a natural design axis. The baseline objective is binary cross-entropy on the primary branch,

$$\mathcal{L}_{\mathrm{BCE}} = -y_t \log\sigma(z_t^{(1)}) - (1-y_t)\log\big(1-\sigma(z_t^{(1)})\big). \tag{6}$$

We explore two alternatives. First, Focal Loss [9] down-weights easy negatives,

$$\mathcal{L}_{\mathrm{focal}} = -\alpha_t\,(1-p_t^*)^\gamma \log(p_t^*), \tag{7}$$

where $p_t^* = p_t$ for a positive target and $1 - p_t$ otherwise, and $\alpha_t = \alpha y_t + (1-\alpha)(1-y_t)$; grid search on validation selects $\gamma = 1.0$ and $\alpha = 0.5$. Second, boundary-aware many-hot supervision [25] expands each ground-truth boundary t_b to its ±1 frame neighborhood,

$$y_{\mathrm{mh}}[t] = \begin{cases} 1, & \exists\, t_b \text{ s.t. } |t - t_b| \le 1, \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

and supervises the auxiliary branch with weight 0.3: $\mathcal{L}_{\mathrm{total}} = \mathcal{L}(\hat{y}^{(1)}, y_{\mathrm{oh}}) + 0.3\,\mathcal{L}(\hat{y}^{(2)}, y_{\mathrm{mh}})$. The small auxiliary weight limits the influence of dense targets on the primary branch's frame-level localization.

These objectives are evaluated as a design space rather than assumed beneficial: the controlled ablation of Section IV-G shows that plain BCE matches both alternatives once post-processing is applied, so the analytic track of this paper anchors on the BCE configuration.

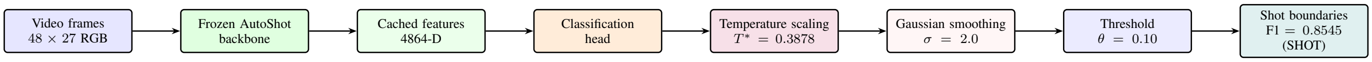

Fig. 1. AutoShotV2 inference pipeline. The searched AutoShot backbone is frozen; AutoShotV2 trains the classification head and applies calibration and temporal smoothing before thresholding. The controlled replication of Section III-I re-fits its own temperature on its validation pool under the frozen validation-only protocol.

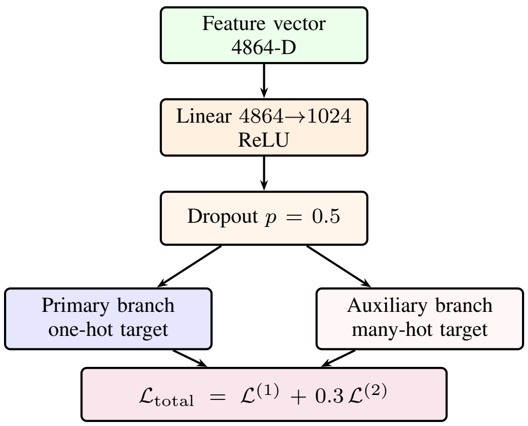

Fig. 2. AutoShotV2 classification head. The primary branch preserves sharp boundary localization; the auxiliary branch can supply denser supervision around each boundary. The controlled ablation selects plain BCE on the primary branch.

TABLE I
DATASET CHARACTERISTICS USED FOR SBD EVALUATION

Property	**BBC**	**ClipShots**	**SHOT**
Content type	Documentary	Open-world video	Short video
Total videos	11	4,039	853
Evaluation videos	11	500	200
Avg. video length	2,945 s	237 s	39.5 s
Avg. shot length	6.57 s	15.34 s	2.59 s
Hard cuts	Many	128,000+	8,890
Gradual transitions	Many	38,000+	2,716
Main challenge	Gradual trans.	Diverse effects	Fast editing

H. Probability Calibration and Temporal Smoothing

SBD decisions rely directly on thresholding model scores, so probability quality matters as much as ranking. Temperature scaling [10] rescales the primary logit without changing its ranking:

$$p_t^{\text{cal}} = \sigma\left(\frac{z_t^{(1)}}{T}\right), \qquad T^* = \arg\min_T \, \mathcal{L}_{\text{BCE}}\left(\sigma\left(\frac{z^{(1)}}{T}\right), y\right), \tag{9}$$

with T^* fitted on validation data. Calibration cannot remove isolated temporal peaks caused by flashes or camera shakes, so a one-dimensional Gaussian filter smooths the calibrated sequence:

$$\tilde{p}_t = \sum_k G_\sigma(k)\, p_{t-k}^{\text{cal}}, \qquad G_\sigma(k) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{k^2}{2\sigma^2}\right). \tag{10}$$

The deployed checkpoint fits $T^* = 0.3878$ on its validation pool; the controlled replication re-fits its own temperature under the frozen protocol below. Both use $\sigma = 2.0$ and $\theta = 0.10$.

I. Validation-Only Protocol Selection

For the controlled track, we select the complete post-processing configuration using five-fold validation. Folds are stratified by source dataset. For each held-out fold, temperature is fitted on the other four folds by minimizing binary cross-entropy. We search $\sigma \in \{0, 1, 2, 3\}$ and $\theta \in \{0.05, 0.10, \ldots, 0.50\}$. Each pair is scored by first computing F1 per source dataset within each held-out fold, averaging those per-source values into a fold-level macro-F1, and then averaging the macro-F1 over the five folds. Ties prefer lower smoothing and then the threshold nearest 0.10. After selecting σ and θ, the final temperature is fitted on all validation videos; the selected values are listed in Table V. The manifest containing folds, key hashes, search space, and selected values is written before test evaluation. We denote the resulting BCE plus temperature plus Gaussian configuration B4.

J. Cache and Run Provenance

The revised pipeline separates a fixed data seed from the training seed. Before feature extraction, it deterministically selects the training-video list and hashes that selected list. A sample cache is reusable only when its schema version, metadata hash, base-checkpoint hash, selected-key hash, video budget, sampling parameters, boundary width, and data seed all match. Each run writes the selected video IDs, per-video sample counts, skipped inputs, configuration, checkpoint hash, and logit-key hash. These identity checks replace the earlier path-tolerant cache check, which recorded the cache size but not the selected video IDs that produced the historical cache.

IV. EXPERIMENTS

A. Datasets and Training Setup

We evaluate on the 200-video SHOT test split [4], all 11 BBC Planet Earth videos [20], and the 500-video ClipShots test split [7]. SHOT is the primary short-video benchmark; BBC tests long documentary footage with many smooth gradual transitions; ClipShots tests open-world transfer. Table I summarizes their characteristics.

AutoShotV2 trains only the classification head with Adam at a constant learning rate; Table II lists the hyperparameters and how each was selected. Performance is measured by precision, recall, and F1, micro-averaged from transition-level TP/FP/FN counts with a matching tolerance of ± 2 frames.

B. Evaluation Tracks and Reproducibility Tiers

The paper separates headline reporting from controlled component analysis. *Track 1 (deployed checkpoint)* provides the headline numbers: the head trained for deployment with

TABLE II
AUTOSHOTV2 HYPERPARAMETERS

Component	Value	Selection
Focal Loss γ	1.0	Grid search
Focal Loss α	0.5	Grid search
Learning rate	10^{-5}	Adam, constant
Weight decay	10^{-4}	Fixed
Batch size	512	Frame-level
Maximum epochs	20	Fixed budget
Dropout	0.5	Fixed
Many-hot weight	0.3	Validation
Gaussian σ	2.0	Validation
Deploy temperature T^*	0.3878	Validation BCE

Note: Focal Loss and many-hot parameters are training-loss options evaluated in the ablation study (Section IV-G); the controlled B4 replication uses binary cross-entropy (BCE) with one-hot labels.

TABLE III
F1 SCORE COMPARISON ACROSS EVALUATION DATASETS

Method	SHOT	BBC	ClipShots
Deep Structured Models (DSM)	–	0.8920	0.7030
TransNet (V1)	–	0.9290	0.7350
TransNetV2, reproduced	–	0.9593	0.7454
AutoShot, reproduced	0.8405	0.9554	**0.7753**
AutoShotV2 (ours), per-dataset oracle best†	**0.8607**	**0.9656**	0.7706

† Peak F1 scores optimized per-dataset under oracle decision thresholds (SHOT and ClipShots evaluated on the supplemental configuration; BBC evaluated on the primary deployment configuration). The supplemental setup uses Focal γ=2.0, α=0.6, lr= 7×10^{-6}, weight decay 10^{-4}, and T^*=0.2413 (reported for potential analysis only; see Section IV-D).

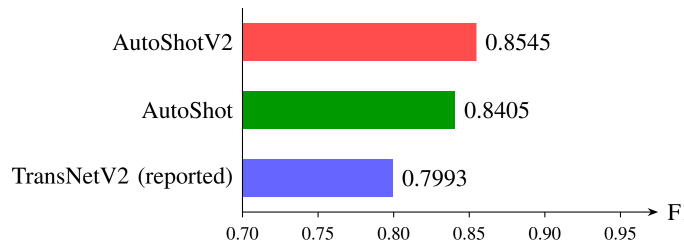

Fig. 3. F1 score comparison on SHOT. AutoShotV2 improves the reproduced AutoShot baseline by 1.40 percentage points and the TransNetV2 (reported) baseline by 5.52 percentage points in F1.

the fixed configuration $T^* = 0.3878$, $\sigma = 2.0$, and $\theta = 0.10$. We regenerated its test logits from the local video splits with full coverage (200/200 SHOT, 11/11 BBC, and 500/500 ClipShots videos) and checksum the resulting logits and result JSON files. *Track 2 (controlled replication)* retrains the head under the frozen validation-only protocol of Section III-I (configuration B4) with full logit caches. Paired bootstrap intervals, calibration diagnostics, ablations, and the seed study are anchored on this controlled track. Because the replication was trained in a separate environment with its own validation pool, its absolute scores differ slightly from the deployed checkpoint. To avoid mixing absolute scales, replication analyses are reported as *differences* against the internal BCE control (A1), while absolute F1 values in the paper refer to the deployed checkpoint.

C. Main Results

Table III reports F1 scores across SHOT, BBC, and ClipShots. Note that this table mixes reported literature baselines and reproduced baselines; only the reproduced rows represent direct pipeline comparisons. Early SBD methods, such as Deep Structured Models (DSM) [7] (F1 of 0.8920 on BBC and 0.7030 on ClipShots) and TransNet (V1) [6] (F1 of 0.9290 on BBC and 0.7350 on ClipShots), report F1 scores that are lower than those of modern architectures, illustrating the steady evolution of spatiotemporal SBD representations.

With the fixed deployment configuration, AutoShotV2 reaches 0.8545 on SHOT and 0.9656 on BBC, corresponding to an improvement of 1.40 percentage points over the reproduced AutoShot baseline on SHOT (Fig. 3). On BBC, the primary deployment configuration achieves an F1 score of 0.9656 (which also represents the best oracle result for this dataset), outperforming the reproduced TransNetV2 baseline (0.9593) by 0.63 percentage points and the reproduced AutoShot baseline (0.9554) by 1.02 percentage points.

On ClipShots, the F1 score of the fixed deployment configuration (0.7529) is higher than the reported scores of DSM by 4.99 percentage points and TransNet (V1) by 1.79 percentage points; it also outperforms the reproduced TransNetV2 baseline by 0.75 percentage points. For cross-domain comparison, a supplemental configuration evaluated with oracle per-dataset thresholds (Table IV) reaches 0.7706 on ClipShots. This closely matches the reproduced AutoShot baseline of 0.7753, which benefits from backbone features tuned directly on the open-world distribution. To distinguish protocol sources, baseline labels indicate whether they are cited from literature or evaluated within our pipeline. Specifically, reproduced baselines are evaluated using our pipeline, whereas reported baselines (for DSM, TransNet, and TransNetV2 on SHOT) are cited from their original publications. Note that the TransNetV2 model was not evaluated on SHOT in the original work; the reported SHOT score of 0.7993 is cited from [4].

D. Supplemental Configuration and Oracle Thresholds

In addition to the primary AutoShotV2 configuration, we conduct an oracle analysis using a supplemental configuration with a revised training and post-processing setup. Specifically, we set $T^* = 0.2413$, $\sigma = 2.0$, Focal Loss parameters $\gamma = 2.0$ and $\alpha = 0.6$, many-hot supervision weight to 0.3, learning rate to 7×10^{-6}, and weight decay to 10^{-4}. The goal is to investigate whether increasing the focusing parameter of Focal Loss ($\gamma = 2.0$ vs. 1.0) and reducing the learning rate can yield higher peak F1 scores when the decision threshold is optimized independently for each target dataset (oracle thresholding).

As shown in Table IV, this supplemental configuration achieves superior peak F1 scores on SHOT (0.8607 vs. 0.8545, a gain of 0.62 percentage points) and ClipShots (0.7706 vs. 0.7529, a gain of 1.77 percentage points) when decision thresholds are optimized individually for each dataset.

Specifically, the "per-dataset oracle best" row in Table III consolidates these peak outcomes by selecting the best F1

TABLE IV
PEAK F1 SCORES UNDER ORACLE ANALYSIS OF THE SUPPLEMENTAL CONFIGURATION

Dataset	Best Threshold	Precision	Recall	Peak F1
SHOT	0.12	0.8408	0.8816	**0.8607**
ClipShots	0.19	0.7011	0.8554	**0.7706**
BBC	0.12	0.9668	0.9628	0.9648

TABLE V
FIXED REPRODUCIBLE EVALUATION PROTOCOL (CONTROLLED TRACK)

Item	Setting
Training objective	BCE, one-hot labels
Training / data seed	42 / 42
Training budget	20 epochs, batch size 512
Sampling	At most 160 frames/video, 3 negatives/positive
Validation size	200 videos
Validation selection	5 source-stratified folds
Selection criterion	Mean macro-F1 across sources/folds
Selected temperature	0.6618
Gaussian smoothing	$\sigma = 2.0$
Decision threshold	$\theta = 0.10$
Matching tolerance	± 2 frames
Bootstrap	10,000 video resamples, seed 42

score per dataset across both the primary and supplemental configurations. This yields 0.8607 on SHOT (from the supplemental configuration), 0.7706 on ClipShots (from the supplemental configuration), and 0.9656 on BBC (from the primary deployment configuration, which outperforms the 0.9648 achieved by the supplemental configuration).

However, this supplemental setup is less robust than the primary configuration under a unified deployment threshold. For example, the best joint threshold for this supplemental configuration ($\theta = 0.17$) yields F1 scores of 0.8518 on SHOT, 0.7581 on ClipShots, and 0.9598 on BBC, resulting in an average F1 of 0.8566. This joint performance is lower than the primary deployment average of 0.8577 across the three datasets. Consequently, while these results reveal domain-specific optimization potential, we retain the primary configuration for the main deployment results of the paper.

E. Controlled Component Analysis

All remaining analyses use Track 2: the controlled B4 replication whose protocol is summarized in Table V.

Relative to the A1 control, the selected B4 configuration improves SHOT by 1.62 percentage points and ClipShots by 4.57 percentage points in F1, while BBC is unchanged after rounding. Temperature scaling alone preserves the validation-selected decision threshold; Gaussian smoothing is the component that changes boundary decisions.

Table VI bootstraps the per-video difference between A1 at its validation-selected threshold and B4 at its frozen configuration (10,000 paired video resamples). The intervals support positive B4 deltas on SHOT and ClipShots. The BBC interval contains zero, so we make no improvement claim for that dataset; BBC also has only 11 sampling units, so its interval should be interpreted cautiously.

TABLE VI
PAIRED VIDEO BOOTSTRAP DIFFERENCE, B4 MINUS A1 (ABSOLUTE PERCENTAGE POINTS)

Dataset	ΔF1 (pp)	95% CI (pp)	Excl. 0
SHOT	+1.62	[+0.93, +2.36]	Yes
BBC	-0.00	[-0.33, +0.41]	No
ClipShots	+4.57	[+2.67, +6.70]	Yes

TABLE VII
FRAME-LEVEL VALIDATION CALIBRATION DIAGNOSTICS

Variant	NLL	Brier	ECE	AECE	B-ECE
Before temperature scaling	0.013006	0.002194	0.007149	0.007149	0.080506
After temperature scaling	0.009591	0.001911	0.002214	0.002243	0.070405

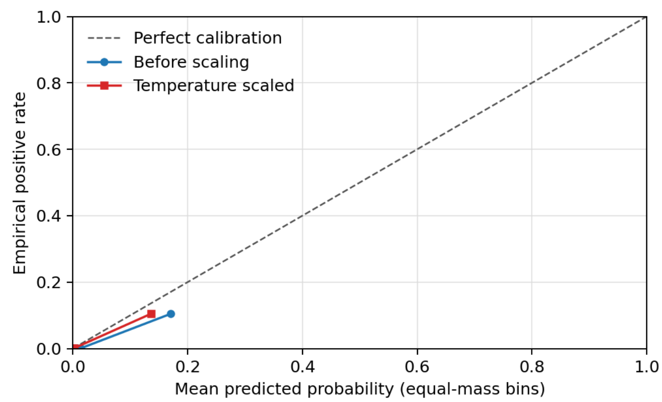

Fig. 4. Reliability diagram on the combined validation split. Temperature scaling reduces aggregate error. Equal-mass bins expose sparsely populated confidence regions.

F. Calibration Diagnostics

We evaluate frame probabilities on the combined 200-video validation split using negative log-likelihood (NLL), Brier score, expected calibration error with 15 equal-width bins (ECE), equal-mass adaptive ECE (AECE), and class-balanced ECE (B-ECE). Table VII shows that temperature scaling improves all diagnostics on this split. The much larger B-ECE reveals that aggregate frame metrics are dominated by negative frames. Because the same validation split fits temperature, these values are used as diagnostics rather than as untouched calibration estimates.

Table VIII reports test-set calibration using the temperature selected on validation. ClipShots has the largest B-ECE despite a small aggregate ECE, consistent with domain-shifted false positives.

G. Controlled Training Ablation

Table IX reports the outcome of the design space of Section III-G as F1 differences against the A1 control. A2, A3,

TABLE VIII
TEST CALIBRATION AT THE VALIDATION-SELECTED TEMPERATURE

Dataset	NLL	Brier	ECE	AECE	B-ECE
SHOT	0.021689	0.004808	0.003014	0.002844	0.133525
BBC	0.013305	0.001662	0.000851	0.001651	0.092471
ClipShots	0.007654	0.001565	0.002520	0.002521	0.156939

TABLE IX
CONTROLLED PHASE 2 ABLATION: ΔF1 AGAINST THE A1 BCE CONTROL (PERCENTAGE POINTS)

Configuration	SHOT	BBC	ClipShots
A2 – Focal only	-0.00	-0.03	-0.17
A3 – Many-hot only	-0.03	-0.08	+0.21
P1 – Gaussian only	+0.54	-1.34	+5.36
P2 – Temperature only	+0.00	+0.00	+0.00
B1 – Focal + many-hot	+0.06	-0.11	+0.23
B4 – Temperature + Gaussian (selected)	**+1.62**	**-0.00**	**+4.57**
B5 – Full candidate	+1.64	-0.19	+4.25

TABLE X
EFFICIENCY AND RUN-ARTIFACT STATISTICS

Item	Value
Frozen backbone parameters	14,299,202
Trainable Phase 2 head parameters	4,983,810
Deployed primary-path parameters	4,982,785
Training videos / sampled frames	400 / 33,174
Training epochs / seed	20 / 42
Post-process CPU cost	21.1 ms / 10^6 frames

and B1 show that Focal Loss and many-hot supervision do not improve the control. B5 combines these training changes with B4 post-processing and is numerically similar on SHOT but slightly weaker on BBC and ClipShots. The observed gain therefore comes from calibrated post-processing rather than from the more complex training objective. A0 is omitted because its per-dataset best-threshold protocol is not a direct controlled comparison.

H. Efficiency and Reproducibility

The backbone remains frozen, so only the Phase 2 head is optimized. Table X reports parameter and run-artifact statistics. The CPU post-processing measurement covers temperature scaling and Gaussian smoothing on cached logits. End-to-end video decoding and backbone inference are excluded. The original run artifact did not retain elapsed training time, so we do not estimate or reconstruct it.

I. ClipShots Transition-Type Recall

We regenerate the ClipShots breakdown directly from the B4 logits and `test.json`. One malformed annotation with reversed endpoints is normalized by sorting the pair. A ground-truth interval is a cut when its width is at most one frame and gradual otherwise. Predictions are class-agnostic, so they are matched once and recall is grouped by the matched ground-truth type. Type-specific precision and F1 are not reported because the model does not predict a transition type.

TABLE XI
CLIPSHOTS RECALL BY GROUND-TRUTH TRANSITION TYPE

Type	GT	Matched	Missed	Recall
Cut	4907	4412	495	0.8991
Gradual	2302	1862	440	0.8089

The gradual-transition recall of 0.8089 is lower than the cut recall of 0.8991, but it is not the dominant source of the overall ClipShots gap. The remaining gap is instead associated with false-positive peaks under open-world effects and domain shift.

J. Training-Seed Robustness

To quantify optimization variability, we retrained the BCE one-hot head 3 times with training seeds 42, 43, and 44 while holding the data seed, the training-video selection, and the sample cache identity fixed. Each run selected its own post-processing configuration with the frozen five-fold validation-only protocol of Section III-I before any test evaluation. Consistent with the delta-only reporting of Section IV-B, we summarize this study by its dispersion: the sample standard deviation of test F1 across seeds is 0.07 percentage points on SHOT, 0.08 percentage points on BBC, and 0.49 percentage points on ClipShots — below 0.5 percentage points on every dataset and an order of magnitude smaller than the paired bootstrap interval widths of Table VI. In these runs, head-training randomness is a smaller source of variation than video sampling. This study isolates training-seed sensitivity end to end, complementing the component analysis of Table IX.

K. Limitations

The study has four important limitations. First, although the headline checkpoint logits have been regenerated from complete local video splits, the three-seed study of Section IV-J measures optimization variability for the controlled pipeline and does not reproduce either checkpoint bit for bit. Second, the historical sample cache recorded 400 videos and 33,174 frames but not the selected IDs; the current manifests fix this for future runs but cannot reconstruct the old cache contents. Third, validation calibration diagnostics reuse the temperature-selection data. Fourth, the frozen representation does not fully control cross-domain false positives on ClipShots. An exploratory experiment that partially unfroze the layer nearest to the classification head reached a ClipShots F1 of 0.7962 but degraded BBC to 0.9341, confirming a multi-domain trade-off that the fully-frozen regime avoids at the cost of lower ClipShots precision. This trade-off is not analyzed further in the present paper, which focuses exclusively on the frozen-feature setting.

V. DISCUSSION

Sources of Performance Gain. The controlled ablation study reframes the training design space presented in Section III-G. Focal Loss and many-hot supervision do not produce consistent practical gains over the BCE control, whereas post-processing changes the operating point: Gaussian smoothing changes boundary decisions, and temperature scaling improves probability quality without changing the ranking. For a frozen-feature pipeline, this ordering is practically important. The post-processing search runs on cached logits and can be repeated for a deployment domain without retraining the representation.

Limitations of Aggregate Calibration Metrics. After temperature scaling, the equal-width ECE falls to 0.002214, yet the class-balanced ECE remains 0.070405. The gap follows from boundary sparsity: non-boundary frames dominate aggregate confidence bins, so a low aggregate score can coexist with poorer calibration on the rare positive class. Reporting only aggregate ECE would therefore overstate calibration quality for SBD and likely for other sparse temporal-event tasks. Event-conditional calibration, such as fitting a separate temperature on boundary-adjacent frames, represents a promising direction for future research, but it requires an independent calibration split that the present protocol does not reserve.

Analysis of Uncertainty Sources. The two uncertainty analyses answer complementary questions. Paired video bootstrap intervals (Table VI) estimate how much the measured deltas depend on which videos are sampled. The three-seed study estimates the sensitivity of the controlled pipeline to head-training randomness. In our runs, the seed-induced standard deviations are below 0.5 percentage points in F1 across all three datasets and are much smaller than the paired bootstrap interval widths. The remaining unquantified term is protocol variability, because the frozen five-fold selection is itself a function of the validation videos.

Precision Bottleneck on ClipShots. The transition-type breakdown shows gradual-transition recall (0.8089) trailing cut recall (0.8991), but the dominant loss is aggregate precision (0.6661): open-world content produces false-positive peaks that survive smoothing. A frozen representation cannot adapt to this distribution shift by construction, so threshold or temperature changes can only trade recall for precision along a fixed curve. Closing the gap likely requires domain-adaptive components, either lightweight adapters on the frozen backbone or domain-specific post-processing selection, rather than further tuning of the present loss-design space.

VI. Conclusion

We presented AutoShotV2, a compact refinement framework for short-video shot boundary detection. The method freezes the NAS-searched AutoShot backbone, trains a lightweight classification head on cached features, and applies temperature scaling and Gaussian smoothing before thresholding. With a single fixed deployment configuration, AutoShotV2 reaches F1 scores of 0.8545 on SHOT, 0.9656 on BBC, and 0.7529 on ClipShots without repeating the architecture search.

The controlled replication clarifies the source of the improvement. Focal Loss and many-hot supervision do not produce consistent practical gains over the BCE control, whereas calibrated post-processing yields paired-bootstrap gains on SHOT and ClipShots. The three-seed study observes F1 standard deviation below 0.5 percentage points on every dataset, and the class-balanced calibration error shows why very small aggregate ECE values should not be interpreted as indicating uniformly calibrated boundary probabilities. Most reported tables are regenerated from tracked result artifacts; controlled analyses include cache identities and frozen selection manifests.

Cross-domain analysis highlights a key remaining challenge: on open-world ClipShots the dominant loss is precision, as visual effects and domain shift create false-positive peaks that survive smoothing. Potential avenues for future work include reserving an independent calibration split, extending calibration to boundary-conditional estimates, and exploring domain-adaptive components such as lightweight adapters on the frozen backbone or per-domain post-processing selection, together with multimodal cues such as audio when visual evidence alone is ambiguous.

Acknowledgment

The authors thank the Faculty of Information Technology, University of Science, Vietnam National University Ho Chi Minh City, for supporting this research.

References

[1] H. Seridi, M. Kurulay, Z. Kouahla, B. Farou, M. N. Kouahla, M. A. Ferrag, and M. A. Boudia, “Shot boundary detection: Fundamental concepts and survey,” in *CEUR Workshop Proceedings*, vol. 2589, 2020.

[2] T. Kar, P. Kanungo, S. N. Mohanty, S. Groppe, and J. Groppe, “Video shot-boundary detection: Issues, challenges and solutions,” *Artificial Intelligence Review*, vol. 57, no. 104, 2024.

[3] S. H. Abdulhussain, “Methods and challenges in shot boundary detection,” *Entropy*, vol. 20, no. 4, p. 214, 2018.

[4] W. Zhu, Y. Yu, W. Huang, J. Huang, W. Feng, X. Wang, X. Yang, C. Hu, C. Liu, B. Fan, and T. He, “AutoShot: A short video dataset and state-of-the-art shot boundary detection,” in *Proc. IEEE/CVF CVPR Workshops*, 2023.

[5] T. Soucek and J. Moravec, “TransNet V2: An effective deep network architecture for fast shot transition detection,” *arXiv preprint arXiv:2008.04838*, 2020.

[6] T. Soucek, J. Moravec, and J. Lokoc, “TransNet: A deep network for fast detection of common shot transitions,” *arXiv preprint arXiv:1906.03363*, 2019.

[7] S. Tang, L. Feng, Z. Kuang, Y. Chen, and W. Zhang, “Fast video shot transition localization with deep structured models,” *arXiv preprint arXiv:1808.04234*, 2018.

[8] T. Elsken, J. H. Metzen, and F. Hutter, “Neural architecture search: A survey,” *Journal of Machine Learning Research*, vol. 20, no. 55, pp. 1–21, 2019.

[9] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, “Focal loss for dense object detection,” in *Proc. IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.

[10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. International Conference on Machine Learning*, 2017, pp. 1321–1330.

[11] C. White, M. Safari, R. Sukthanker, B. Ru, T. Elsken, A. Zela, D. Dey, and F. Hutter, “Neural architecture search: Insights from 1000 papers,” *arXiv preprint arXiv:2301.08727*, 2023.

[12] P. Ren, Y. Xiao, X. Chang, P.-Y. Huang, Z. Li, X. Chen, and X. Wang, “A comprehensive survey of neural architecture search: Challenges and solutions,” *arXiv preprint arXiv:2006.02903*, 2020.

[13] A. Esteve et al., “Shot boundary detection with 3D depthwise convolutions and visual attention,” *Sensors*, vol. 23, no. 12, p. 5620, 2023.

[14] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, “Learning spatiotemporal features with 3D convolutional networks,” in *Proc. IEEE International Conference on Computer Vision*, 2015, pp. 4489–4497.

[15] J. C. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” in *Advances in Large Margin Classifiers*. Cambridge, MA, USA: MIT Press, 1999, pp. 61–74.

[16] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 694–699.

[17] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, “Measuring calibration in deep learning,” in *Proc. IEEE/CVF CVPR Workshops*, 2019.

[18] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for NLP,” in *Proc. International Conference on Machine Learning*, 2019, pp. 2790–2799.

[19] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. International Conference on Learning Representations*, 2022.

[20] L. Baraldi, C. Grana, and R. Cucchiara, “Shot and scene detection via hierarchical clustering for re-using broadcast video,” in *Proc. Computer Analysis of Images and Patterns*, 2015, pp. 801–811.

[21] A. Hassanien, M. Elgharib, A. Selim, S.-H. Bae, M. Hefeeda, and W. Matusik, “Large-scale, fast and accurate shot boundary detection through spatio-temporal convolutional neural networks,” *arXiv preprint arXiv:1705.03281*, 2017.

[22] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Proc. Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.

[23] Z. Guo, X. Zhang, H. Mu, W. Heng, Z. Liu, Y. Wei, and J. Sun, “Single path one-shot neural architecture search with uniform sampling,” in *Proc. European Conference on Computer Vision*, 2020, pp. 544–560.

[24] H. Liu, K. Simonyan, and Y. Yang, “DARTS: Differentiable architecture search,” in *Proc. International Conference on Learning Representations*, 2019.

[25] R. Müller, S. Kornblith, and G. E. Hinton, “When does label smoothing help?” in *Proc. Advances in Neural Information Processing Systems*, 2019, pp. 4694–4703.